

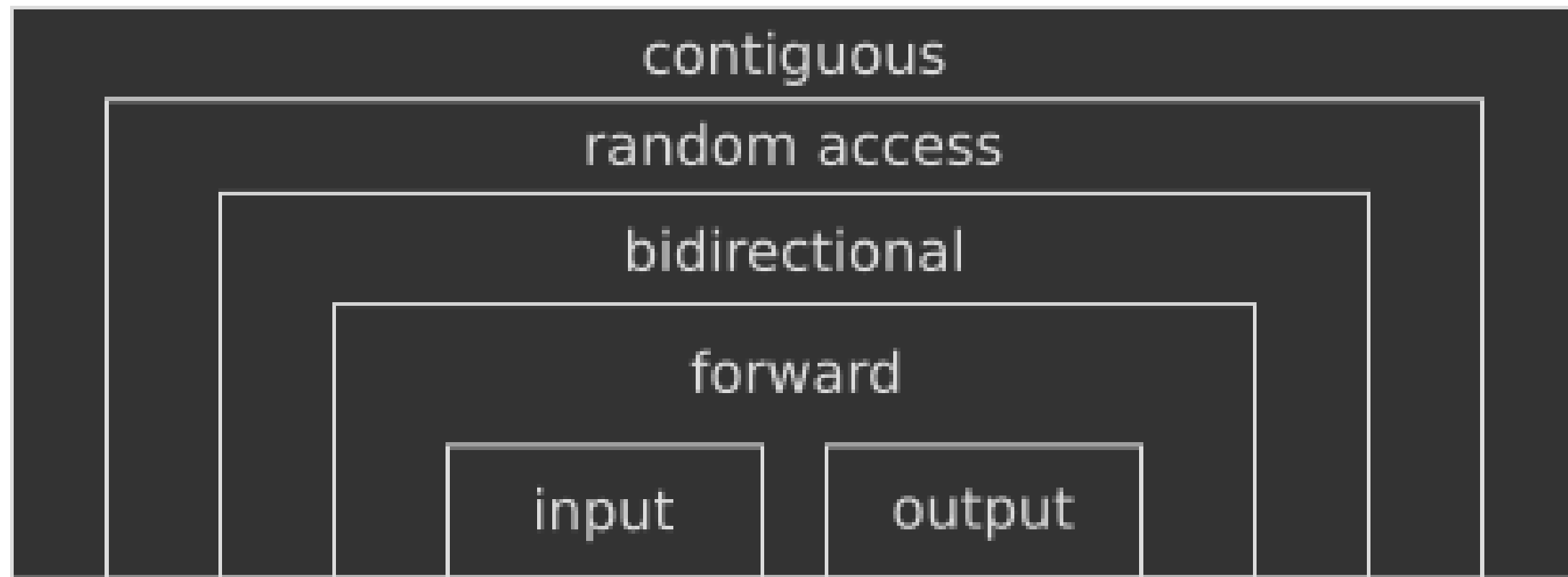
# Семинар №2

## Итераторы. Начало

# Итераторы

- Обобщение указателей
- Можно получить с помощью методов `begin/end` и т.д.
- `based-range for`

# Иерархия итераторов



# Требования к видам итераторов

| Iterator category                                                                                     | Operations and storage requirement |          |                         |                      |           |               |                    |
|-------------------------------------------------------------------------------------------------------|------------------------------------|----------|-------------------------|----------------------|-----------|---------------|--------------------|
|                                                                                                       | write                              | read     | increment               |                      | decrement | random access | contiguous storage |
|                                                                                                       |                                    |          | without multiple passes | with multiple passes |           |               |                    |
| <i>LegacyOutputIterator</i>                                                                           | Required                           |          | Required                |                      |           |               |                    |
| <i>LegacyInputIterator</i><br>(mutable if supports write operation)                                   |                                    | Required | Required                |                      |           |               |                    |
| <i>LegacyForwardIterator</i><br>(also satisfies <i>LegacyInputIterator</i> )                          |                                    | Required | Required                | Required             |           |               |                    |
| <i>LegacyBidirectionalIterator</i><br>(also satisfies <i>LegacyForwardIterator</i> )                  |                                    | Required | Required                | Required             | Required  |               |                    |
| <i>LegacyRandomAccessIterator</i><br>(also satisfies <i>LegacyBidirectionalIterator</i> )             |                                    | Required | Required                | Required             | Required  | Required      |                    |
| <i>LegacyContiguousIterator</i> <sup>[1]</sup><br>(also satisfies <i>LegacyRandomAccessIterator</i> ) |                                    | Required | Required                | Required             | Required  | Required      | Required           |

# Функции, работающие с итераторами

1. find, find\_if, find\_if\_not; count, count\_if
2. all\_of, any\_of, for\_each
3. remove, remove\_if, replace, replace\_if
4. sort, stable\_sort, partition
5. is\_permutation, next\_permutation, is\_heap, make\_heap
6. rotate, reverse

# Задание:

- Реализуйте функции из слайда выше (может, кроме строк 4, 5). Напишите тесты, сравните результат с функциями из стандартной библиотеки. Запустите под санитайзерами.

# Какие итераторы нужны? Input или Forward?

- <https://leetcode.com/problems/best-time-to-buy-and-sell-stock/>
- <https://leetcode.com/problems/best-time-to-buy-and-sell-stock-ii/>
- <https://leetcode.com/problems/majority-element/>
- <https://leetcode.com/problems/container-with-most-water/>
- <https://leetcode.com/problems/trapping-rain-water/>